BIOT 6900 — Module 2, Part 3

Alzheimer's Disease Multi-Omics Integration

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Assignment: Required Module 2 assignment — gene-level integration of Alzheimer's transcriptomic, proteomic, and genomic evidence

Summary of pipeline and results

I integrated three independently sourced Alzheimer's disease datasets at the gene level:

- Transcriptomics: GEO2R differential expression analysis of GSE5281 (hippocampus, 13 control vs. 10 AD samples), giving log2 fold change and adjusted p-values per probe.
- Proteomics: Supplementary Table 2 from Johnson et al. (2022, Nature Neuroscience), a TMT mass-spectrometry protein co-expression study of over 500 brain tissues, giving an AD-vs-control abundance difference and Holm-adjusted p-value per protein.
- Genomics: The GWAS Catalog, filtered to Alzheimer's disease associations, giving mapped gene and association p-value per SNP.

None of these three studies share patient samples, so no patient-level correlation was possible. Instead, following the required workflow, I harmonized identifiers on gene symbol, collapsed each source to one row per gene (keeping the most significant p-value when a gene had multiple entries), and inner-joined all three tables. This left 731 genes with usable evidence in all three layers — the reality check for how much signal survives a strict three-way, cross-cohort join.

For each surviving gene I computed a boolean concordant field (whether the RNA and protein fold-change point in the same direction — 347 of 731 genes, 47 percent) and a score (an equal-weighted, per-layer magnitude score combining absolute RNA log2 fold change, absolute protein log2 fold change, and negative log10 of the GWAS p-value, each weighted one third).

The top-ranked genes by this score, exported to targets_ad.csv, were:

Gene	RNA log2FC	Protein log2FC	Concordant	Score
APOE	1.170	0.147	True	103.005
APOC1	-1.040	-0.093	True	102.944
TOMM40	0.063	-0.049	False	98.371
BCAM	-1.560	-0.029	True	64.248
ALDH1A2	-0.200	-0.112	True	27.903
CLPTM1	-0.145	-0.043	True	21.195
CLU	-0.918	0.139	False	16.019
FARP2	-1.820	0.036	False	14.518
SAMHD1	-3.650	0.054	False	13.134
ABCA7	-0.131	-0.036	True	12.755

1. Weighting

I used equal weights (one third transcriptomic, one third proteomic, one third genomic) for the multi-evidence score. I chose this deliberately rather than defaulting to it out of convenience.

The argument for equal weighting is that each layer captures a genuinely different kind of evidence — GWAS captures germline, causal risk; transcriptomics and proteomics capture downstream disease-state changes in affected tissue. Since the assignment's goal is to surface genes with converging, multi-modal support rather than to model a specific causal chain, treating the three lines of evidence as equally informative avoids implicitly asserting a mechanism I haven't actually tested.

The strongest counter-argument is that GWAS p-values and expression fold-changes are not naturally on the same scale or with the same statistical meaning — a GWAS p-value reflects population-level association strength across potentially hundreds of thousands of individuals, while the RNA and protein fold-changes here come from far smaller cohorts (23 samples for RNA, roughly 500 tissues for protein) and reflect magnitude of change, not significance. A more principled version of this pipeline might up-weight transcriptomics and proteomics precisely because they are closer to the phenotype being studied, and treat GWAS as a prior or filter rather than an equal partner in the same additive score. I did not implement that here, both to keep the pipeline consistent with the required workflow and because there is no single defensible correct weighting without a specific downstream use case in mind.

2. Top targets

The pipeline recovered APOE as the single top-ranked gene, which is the expected and correct outcome — APOE is the strongest known genetic risk factor for late-onset Alzheimer's disease, and its recovery here is a direct validation that the pipeline is working correctly rather than producing noise.

Notably, APOC1 and TOMM40 also rank in the top three. This is not three independent discoveries — APOE, APOC1, and TOMM40 sit within roughly 20 kb of each other on chromosome 19q13.32, in a region of strong linkage disequilibrium. GWAS studies routinely report significant associations at all three genes for this reason, even though the causal signal is very likely driven by APOE alone (specifically the epsilon-4 allele). This is a useful example of why gene-level GWAS hits need biological interpretation rather than being taken as independently causal.

Of the classic AD genes named in the assignment (APOE, TREM2, BIN1, CLU, PICALM), I recovered APOE and CLU in the top ten; TREM2, BIN1, and PICALM were not in the top fifteen, likely because they did not clear the three-way join — that is, they may be missing usable data in one of the three source layers, most plausibly the small GEO2R RNA dataset or gaps in the mass-spec proteomics coverage.

A non-obvious hit worth a second look: ABCA7 (rank ten, concordant). ABCA7 is a genuine, well-replicated AD risk gene involved in lipid transport and amyloid-beta clearance, but it is less universally cited than APOE, CLU, or PICALM in introductory material, so it is a good example of the pipeline surfacing a real, literature-supported target beyond the most famous names.

3. Reading a discordant gene

I chose CLU (clusterin) as the discordant gene to examine: RNA log2 fold change is -0.918 (lower in AD), but protein log2 fold change is +0.139 (slightly higher in AD).

CLU encodes clusterin, a secreted extracellular chaperone protein strongly implicated in amyloid-beta clearance and a well-established AD GWAS hit in its own right. A plausible biological explanation for the RNA-down, protein-up pattern is that clusterin is heavily regulated post-transcriptionally and at the level of secretion and extracellular stability, rather than purely at the transcript level — clusterin protein can accumulate extracellularly, including in senile plaques, even when its transcription in surviving neurons or glia is reduced, and protein turnover, secretion rate, and complex formation with amyloid-beta can all decouple protein abundance from mRNA levels. It is also worth noting that the RNA and protein measurements come from different cohorts (GSE5281 versus the ROSMAP and Banner TMT proteomics cohort) and different specific brain regions and cell-type compositions, so some of this discordance may reflect real cohort or tissue heterogeneity rather than a single unified biological mechanism, a limitation discussed further below.

4. Limitation

This entire integration was performed at the gene level, matching genes by symbol across three cohorts that share no common patients: the RNA data comes from GSE5281 (hippocampus, AD vs. control), the protein data from Johnson et al.'s ROSMAP and Banner TMT cohort (dorsolateral prefrontal cortex), and the genomic data from the aggregated GWAS Catalog (thousands of individuals across many studies). Because there is no shared sample identifier across these sources, I cannot correlate any individual patient's RNA level against their own protein level or genetic risk — a gene can only be said to show fold-change in RNA in one population and fold-change in protein in a different population, which is a fundamentally weaker claim than saying this patient's high RNA corresponds to their high protein.

This stands in direct contrast to the matched CPTAC breast cancer case from Part 1, where RNA, protein, and mutation data all came from the same 122 tumors, allowing genuine patient-level, matched multi-omic correlation, including a real RNA-protein correlation coefficient. Here, any apparent concordance or discordance is a comparison of population-level average effects computed on non-overlapping cohorts and even different brain regions, not a per-patient biological relationship. Conclusions from this pipeline should therefore be read as these genes show convergent evidence for AD across independent studies, not as claims about how a given gene behaves within a specific patient's tissue.

Data quality note

One data-cleaning step is worth flagging: several GWAS Catalog p-values were recorded as exactly zero, a floating-point underflow artifact rather than a true value of zero. I floored these to the smallest nonzero p-value observed in the dataset before taking the negative log base ten, to avoid infinite genomic scores. This is a standard, defensible handling of underflowed p-values, but it does mean the exact genomic sub-score for those genes is a conservative estimate rather than the true, unrepresentable value.